Al Muqshith Shifan

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SUMMARY

MSc Computer Science student researching reinforcement learning (RL) and LLM agents for games, and **first author of an IEEE CoG 2026 paper**. Builds deep RL pipelines in PyTorch, Unity ML-Agents tools, and LLM benchmarks. CCNA certified.

EDUCATION

MSc, Computer Science (Software Design, AI/RL focus), Ontario Tech University — May 2026 - Present
Bachelor of IT, Networking and Cybersecurity, Ontario Tech University | GPA 3.8/4.3 (final 2 years), Dean's List — Jun 2026

PUBLICATIONS

AL Muqshith Mohammed Shifan, Kevin Christopher Chua, Ali Neshati, Loutfouz Zaman, and Cristiano Politowski. "RIDGE: State-Conditioned Reward Blending for Behavioral Coverage in Deep RL Game Agents." *IEEE Conference on Games (CoG)*, Madrid, Spain, Sept. 1-5, 2026. **Short paper, first author, presented.** Preprint: arXiv:2611.05601.

- One RL agent blends 4 personas via a **multi-head PPO critic**; on Crafter it learned crafting and combat, which no baseline did.

AL Muqshith Mohammed Shifan (second author). "The Resting Wrist as a Two-Axis Input Channel: Decoding Surface-Anchored Strokes from a Commodity Smartwatch IMU." *ACM CHI 2027*. **Under review.**

- Decoded finger strokes from a smartwatch IMU (16 participants); personalization raised 8-way accuracy from 54% to **74%**.

EXPERIENCE

Graduate Research Assistant, Ontario Tech University — May 2026 - Present

- Member of **Code & Sorcery Lab (Empirical Game Research)**, GaMER Lab, and UXID Lab.
- Built RIDGE (CoG 2026), ROGUE, and AFTERSTATE, and co-developed PEAK: RL playtesting agents and LLM characters.

Teaching Assistant, Ontario Tech University — Sep 2025 - Present

- Web Development: assisted students in weekly labs and graded coursework for all 60 students.
- Advanced Networking I: taught 50+ students; also supported Networking I and Advanced Networking II and III.

Network Administrator (Summer Research Assistant), Ontario Tech University — Jun 2025 - Sep 2025

- Wrote the lab's first working **Intel Tofino P4** telemetry programs; fixed Linux driver faults, cutting server crashes by 30%.

Cyber Security Analyst, YYC Beeswax Ltd | Remote — Jun 2025 - Sep 2025

- Found 15 exploitable issues in a WooCommerce and Stripe stack and wrote the team's pre-deployment hardening checklist.

PROJECTS

ROGUE: Persona-Based RL for Any Unity Game | *Python, PyTorch, C#, Unity, ML-Agents* — 2026

- One-file C# SDK adds persona-based RL to any Unity game. Tested on 5 games: **1.6** Obstacle Tower floors vs. 0.66 random.

AFTERSTATE: Emotional Memory Benchmark for LLM Characters (MSc thesis, in progress) | *Python, LLMs* — 2026

- Measures how much emotion LLM characters lose when they compress their memory; 60 stories, 8 facts and 8 feelings each.

RAZE: Zero-Parameter Agent for the TinyBrains Ants AI Competition | *Python, ONNX* — 2026

- Hand-built ONNX network with no learned weights, under 16 KB; **ranked 1st of 69** on the Open ladder and 1st in 3 size classes.

PEAK: Platformer Engine for RL Benchmarking | *Python, Gymnasium, Stable-Baselines3, Hydra* — 2025 - 2026

- Co-developed a deterministic DRL benchmark with 11 stages; spatial-hash collision runs at **1000+ steps per second**.

Gestura: Gesture-Based Medical Screening (HackHive 2026) | *Python, Flask, MediaPipe, Three.js, Gemini* — 2026

- Non-verbal patient intake: real-time hand tracking selects pain regions on a 3D body model; Gemini writes clinician notes.

SKILLS AND CERTIFICATIONS

Languages: Python, C++, C#, SQL, Bash, JavaScript, HTML/CSS, P4
ML / AI: PyTorch, Stable-Baselines3, Gymnasium, Unity ML-Agents, PPO, ONNX, LLM evaluation, Gemini API, MediaPipe, NumPy
Tools and Networking: Git, Docker, Flask, Linux, Cisco IOS, Intel Tofino P4, Wireshark, Nmap, Vulnerability Assessment
Certifications: Cisco Certified Network Associate (CCNA), 2024 - 2027 | TryHackMe, 200+ hours